

MATH 162, Winter 2026
SCRIPT 11: Limits and Continuity

Throughout this sheet, we let $f, g: A \rightarrow \mathbb{R}$ be real-valued functions with domain $A \subset \mathbb{R}$, unless otherwise specified.

Definition 11.1. Let $a \in LP(A) \subset \mathbb{R}$. A *limit* of f at a is a number $L \in \mathbb{R}$ satisfying the following condition: for every $\epsilon > 0$, there exists a $\delta > 0$ such that

$$\text{if } x \in A \text{ and } 0 < |x - a| < \delta, \quad \text{then } |f(x) - L| < \epsilon.$$

(See *Additional Exercise 2* for one-sided limits.)

Lemma 11.2. *Limits are unique: if L and L' are both limits of f at a point a , then $L = L'$.*

Proof of Lemma 11.2. We want to show that limits are unique. Let $L, L' \in \mathbb{R}$ such that they are both the limits of $a \in \mathbb{R}$. We want to show that $L = L'$.

Suppose for the sake of contradiction that $L \neq L'$. Then, $\forall \epsilon > 0, \exists \delta, \delta' > 0$ such that if $x \in A$ and $0 < |x - a| < \delta$ and $0 < |x - a| < \delta'$ then

$$|f(x) - L| < \epsilon \quad |f(x) - L'| < \epsilon$$

Let $\epsilon = \frac{|L - L'|}{2}$. Then, we have that

$$|f(x) - L| < \frac{|L - L'|}{2} \quad |f(x) - L'| < \frac{|L - L'|}{2}$$

Now, let $\delta_0 = \min(\delta, \delta')$. Then, we have that if $0 < |x - a| < \delta_0$ then, by the Triangle Inequality, we have that

$$\begin{aligned} |L - L'| &= |(L - f(x)) + (f(x) - L')| \\ |L - L'| &\leq |f(x) - L| + |f(x) - L'| < |L - L'| \end{aligned}$$

This contradicts the principle of trichotomy and thereby completes the proof. □

Definition 11.3. If L is the limit of f at a , we write

$$\lim_{x \rightarrow a} f(x) = L.$$

Exercise 11.4. Give an example of a set $A \subset \mathbb{R}$, a function $f: A \rightarrow \mathbb{R}$, and a point $a \in LP(A)$ such that $\lim_{x \rightarrow a} f(x)$ does not exist.

Proof of Exercise 11.4. Let $A = (0, 1) \cup (1, 2)$ and let $f : A \rightarrow \mathbb{R}$ be defined as $f(x) = 1$ if $x \in (0, 1)$ and $f(x) = -1$ if $x \in (1, 2)$. Consider the point $1 \in LP(A)$. We want to show that $\lim_{x \rightarrow 1} f(x)$ does not exist. Suppose for the sake of contradiction that L is a limit of $f(x)$ at $x = 1$.

Case 1: $L = 1$. Then, take $\epsilon = 0.5$. We have that for all $\delta > 0$, there exists $x \in (1, 1 + \delta) \cap (1, 2)$. Then, we have that $f(x) = -1$ and therefore $|f(x) - L| = |-1 - 1| = 2 \not< 0.5 = \epsilon$, a contradiction.

Case 2: $L \neq 1$. Then, take $\epsilon = \frac{|1-L|}{2}$. We have that for all $\delta > 0$, there exists $x \in (1 - \delta, 1) \cap (0, 1)$. Then, we have that $f(x) = 1$ and therefore $|f(x) - L| = |1 - L| \not< \frac{|1-L|}{2} = \epsilon$, a contradiction.

In both cases we reach a contradiction, and therefore $\lim_{x \rightarrow 1} f(x)$ does not exist. $\square$

We now deduce a more concrete characterization of continuity at a point (Definition 9.9).

Theorem 11.5. *Let $x \in A$. Then the following are equivalent:*

- (i) f is continuous at x .
- (ii) For every $\epsilon > 0$, there exists $\delta > 0$ such that

$$\text{if } y \in A \text{ and } |y - x| < \delta, \quad \text{then } |f(y) - f(x)| < \epsilon.$$

- (iii) Either $x \notin LP(A)$ or $\lim_{y \rightarrow x} f(y) = f(x)$.

Proof of Theorem 11.5 (i.) $\rightarrow$ (ii.) Let f be continuous at x . We want to show that For every $\epsilon > 0$, there exists $\delta > 0$ such that if $y \in A$ and $|y - x| < \delta$, then $|f(y) - f(x)| < \epsilon$.

As f is continuous at x , we have that there exists an open set $S \ni x$. Now, let $\epsilon > 0$ and $R = (f(x) - \epsilon, f(x) + \epsilon)$. As we have that f is continuous at x , we have that $S \cap A \subset f^{-1}(R)$. We can write this as $f(S \cap A) \subset R$.

Now, consider the open set S . As S is open we have that $\forall x \in S$ there exists a $\delta > 0$ such that $(x - \delta, x + \delta) \subset S$. By definition, as $0 < |y - x| < \delta$, we may say that $y \in (x - \delta, x + \delta)$. From this we get that $y \in S \cap A$ and $f(y) \in f(S \cap A)$. As $f(S \cap A) \subset R$, we have that $f(y) \in (f(x) - \epsilon, f(x) + \epsilon)$. We can write this as $|f(y) - f(x)| < \epsilon$. $\square$

Proof of 11.5 (ii.) $\rightarrow$ (iii.) Let $x \in A$ such that for every $\epsilon > 0$, there exists $\delta > 0$ such that if $y \in A$ and $|y - x| < \delta$, then $|f(y) - f(x)| < \epsilon$.

Firstly, suppose that $x \notin LP(A)$. Then, this is proven by definition.

Secondly, suppose that $x \in LP(A)$ and pick $\epsilon > 0$. Then, for every region $R \ni x$, $A \cap R \setminus \{x\} \neq \emptyset$. Therefore, we can write R as $R = (x - \delta, x + \delta)$ and $y \in (x - \delta, x + \delta)$. Then, by definition

$$y \in A \cap R \setminus \{x\}$$

Now, consider $y \in (x - \delta, x + \delta)$. We can write this as $|y - x| < \delta$. Then, as $y \in (x - \delta, x + \delta)$ and $y \in A$ we have that $|f(y) - f(x)| < \epsilon$. Therefore, we can write that as $\forall \epsilon > 0, \exists \delta > 0$ where $y \in A$ and $0 < |y - x| < \delta$ and therefore $|f(y) - f(x)| < \epsilon$ where $f(x)$ is the limit. Thus, this works to prove $\lim_{y \rightarrow x} f(y) = f(x)$.

This suffices to prove that either $x \notin LP(A)$ or $\lim_{y \rightarrow x} f(y) = f(x)$. □

Proof of Theorem 11.5 (iii.) $\rightarrow$ (i.) We want to show that if either $x \notin LP(A)$ or $\lim_{y \rightarrow x} f(y) = f(x)$ then f is continuous at x .

Firstly, suppose that $x \notin LP(A)$ and $x \in A$. As $x \notin LP(A)$, we can say that for any open set S containing x , we have that $A \cap S \setminus \{x\} = \emptyset$. Therefore, we have that $S \cap A = \{x\}$. Now, consider any region R that contains $f(x)$. To show that f is continuous at x , it suffices to show that $f(S \cap A) \subset R$. As $S \cap A = \{x\}$, we have that $f(S \cap A) = f(x)$, and therefore $f(S \cap A) \subset R$. Therefore, we have that f is continuous at x .

Secondly, suppose that $\lim_{y \rightarrow x} f(y) = f(x)$. Now, consider any region R such that $f(x) \in R$. As R is open, we can pick $\epsilon > 0$ such that $(f(x) - \epsilon, f(x) + \epsilon) \subset R$. Then, as $\lim_{y \rightarrow x} f(y) = f(x)$, there exists $\delta > 0$ such that if $y \in A$ and $|y - x| < \delta$ then $|f(y) - f(x)| < \epsilon$. Now, consider the open set $V = (x - \delta, x + \delta)$. Then, we have that $y \in V \cap A$ and therefore $f(y) \in (f(x) - \epsilon, f(x) + \epsilon) \subset R$. Therefore, we have that $f(V \cap A) \subset R$, which suffices to prove that f is continuous at x .

This suffices to prove that if either $x \notin LP(A)$ or $\lim_{y \rightarrow x} f(y) = f(x)$ then f is continuous at x . □

Exercise 11.6. (a) Let $a, b \in \mathbb{R}$ and let $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = ax + b$. Show that f is continuous at every $x \in \mathbb{R}$.

(b) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = \begin{cases} 1 & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$ Show that f is not continuous at 0.

Exercise 11.6 (a). Let $a, b \in \mathbb{R}$ and let $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = ax + b$. We want to show that f is continuous at every $x \in \mathbb{R}$. For this, it suffices to show that for every $\epsilon > 0$ there exists $\delta > 0$ such that if $y \in \mathbb{R}$ and $|y - x| < \delta$ then $|f(y) - f(x)| < \epsilon$.

Let $y \in \mathbb{R}$. We have that $|f(y) - f(x)|$ can be written as $|(ay + b) - (ax + b)| = |ay - ax|$. This can be simplified to $|a|(y - x) = |a||y - x|$.

Now, let $\delta = \frac{\epsilon}{|a|+1}$. Then, we can write $|f(y) - f(x)| = |a||y - x|$ as $|f(y) - f(x)| < a \cdot \delta$. Thus, we get that $|f(y) - f(x)| < |a| \cdot \frac{\epsilon}{|a|+1} < \epsilon$. As this works applies for all $x \in \mathbb{R}$, we have that f is continuous at every $x \in \mathbb{R}$. $\square$

Exercise 11.6 (b). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = \begin{cases} 1 & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$. We want to show that f is not continuous at 0.

Let $0 \in \mathbb{R}$ and let $\epsilon > 0$. To show that f is not continuous at 0, it suffices to show that there exists some $\epsilon > 0$ such there is not a $\delta > 0$ where ...

Suppose for sake of contradiction that $\lim_{x \rightarrow 0} f(x) = 0 = f(0)$. Consider $\epsilon = 0.5$ and $y = \frac{\delta}{2}$, then $|y - 0| = \frac{\delta}{2} < \delta$ but $|f(y) - f(0)| = |1 - 0| = 1 \not< 0.5 = \epsilon$. This completes the proof. $\square$

Exercise 11.7. Show that the absolute value function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = |x|$ is continuous.

Exercise 11.7. We want to show that the absolute value function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = |x|$ is continuous. For this, it suffices to show that f is continuous at an arbitrary point $x \in \mathbb{R}$.

Let $x_0 \in \mathbb{R}$. To show that f is continuous at x_0 , it suffices to show that $x_0 \notin LP(A)$ or $\lim_{y \rightarrow x} f(y) = f(x_0)$. By definition, we have that $x \in LP(A)$ and therefore we want to show that $\lim_{y \rightarrow x} f(y) = f(x_0)$.

We want to show that for every $\epsilon > 0$, there exists $\delta > 0$ such that if $y \in \mathbb{R}$ and $0 < |y - x_0| < \delta$ then $|f(y) - f(x_0)| < \epsilon$. Let $y \in \mathbb{R}$ such that $y \neq x_0$. Also, let $\delta = \frac{\epsilon}{2}$. Then, by definition of the function, we have that $|f(y) - f(x_0)| = |y - x_0| < \epsilon$. As $\delta = \frac{\epsilon}{2}$, we have that $|y - x_0| < \frac{\epsilon}{2} < \epsilon$. This works to complete the proof. $\square$

Given real-valued functions f and g , we define new functions $f + g$, fg and $\frac{1}{f}$ by

- $(f + g)(x) = f(x) + g(x)$;
- $(fg)(x) = f(x) \cdot g(x)$; and
- $\frac{1}{f}(x) = \frac{1}{f(x)}$, provided that $f(x) \neq 0$.

We wish to understand the limits of $f + g$, fg and $\frac{1}{f}$ in terms of the limits of f and g .

Lemma 11.8. *If $\lim_{x \rightarrow a} f(x) = L > 0$ then there exists a $\delta > 0$ such that $f(x) > \frac{L}{2}$ for all $x \in A \cap [(a - \delta, a + \delta) \setminus \{a\}]$. The analogous statement is true if $\lim_{x \rightarrow a} f(x) = L < 0$.*

Proof of Lemma 11.8. Let $\lim_{x \rightarrow a} f(x) = L$ such that $L > 0$. We want to show that there exists $\delta > 0$ such that $f(x) > \frac{L}{2}$ for all $x \in A \cap [(a - \delta, a + \delta) \setminus \{a\}]$.

Let $\lim_{x \rightarrow a} f(x) = L$. Then, we have that for all $\epsilon > 0$, there exists $\delta > 0$ such that if $x \in A$ and $|x - a| < \delta$ then $|f(x) - L| < \epsilon$. We can rewrite $|x - a| < \delta$ as $-\delta < x - a < \delta$ or, alternatively, $-\delta + a < x < \delta + a$. Thus, we have that $x \in (a - \delta, a + \delta)$. As $a, x \in A$, this is equivalent to $x \in A \cap [(a - \delta, a + \delta) \setminus \{a\}]$. As $|x - a| \neq 0$, we have that $x \neq a$ and therefore, we get that $x \in A \cap [(a - \delta, a + \delta) \setminus \{a\}]$.

Now, choose $\epsilon = \frac{L}{2}$. Then, we have that $|f(x) - L| < \frac{L}{2}$. Now, we have that $\forall x \in A \cap [(a - \delta, a + \delta) \setminus \{a\}]$, $f(x)$ must be greater than $\frac{L}{2}$ for $|f(x) - L| < \frac{L}{2}$. This suffices to prove the statement.

For the analogous statement, let $L < 0$. Choose $\epsilon = \frac{|L|}{2}$. Then, we have that $|f(x) - L| < \frac{|L|}{2}$, which gives us $f(x) < \frac{L}{2} < 0$ for all $x \in A \cap [(a - \delta, a + \delta) \setminus \{a\}]$. This suffices to prove the analogous statement. □

Theorem 11.9. *Suppose that $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$. Then,*

- (a) $\lim_{x \rightarrow a} (f + g)(x) = L + M$.
- (b) $\lim_{x \rightarrow a} (fg)(x) = L \cdot M$.
- (c) *Suppose that $\lim_{x \rightarrow a} f(x) = L \neq 0$. Then $\lim_{x \rightarrow a} \frac{1}{f}(x) = \frac{1}{L}$.*

Proof of Theorem 11.9(a.) Let $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$. Then, suppose there exists a function $h(x) = (f + g)(x)$. To show that $\lim_{x \rightarrow a} (f + g)(x) = L + M$, it suffices to show that $|h(x) - (L + M)| < \epsilon$.

Now, as $\lim_{x \rightarrow a} f(x) = L$, we have that for every $\epsilon > 0$, there exists $\delta_1 > 0$ such that if $x \in A$ and $|x - a| < \delta_1$ then $|f(x) - L| < \epsilon$. Similarly, we have that for every $\epsilon > 0$, there exists $\delta_2 > 0$ such that if $x \in A$ and $|x - a| < \delta_2$ then $|g(x) - M| < \epsilon$.

Now, as this statement applies for all $\epsilon > 0$, consider $\frac{\epsilon}{2}$. Then, by definition, we have that there exists $\delta_1 > 0$ such that if $x \in A$ and $|x - a| < \delta_1$ then $|f(x) - L| < \frac{\epsilon}{2}$ and there exists $\delta_2 > 0$ such that if $x \in A$ and $|x - a| < \delta_2$ then $|g(x) - M| < \frac{\epsilon}{2}$. Then, we have that

$$|f(x) - L| + |g(x) - M| < \frac{\epsilon}{2} + \frac{\epsilon}{2}$$

By **Lemma 8.9(a)**, we can rewrite this as

$$|(f + g)(x) - (L + M)| \leq |f(x) - L| + |g(x) - M| < \epsilon$$

Then, as $h(x) = (f + g)(x)$, we have that $|h(x) - (L + M)| < \epsilon$. This suffices to prove that $\lim_{x \rightarrow a} (f + g)(x) = L + M$. $\square$

Proof of Theorem 11.9(b). Let $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$. Then, suppose there exists a function $h(x) = (fg)(x)$. To show that $\lim_{x \rightarrow a} (fg)(x) = L \cdot M$, it suffices to show that $|h(x) - (L \cdot M)| < \epsilon$.

As $\lim_{x \rightarrow a} f(x) = L$, taking desired precision $\epsilon_0 = 1$, there exists $\delta_1 > 0$ such that if $x \in A$ and $|x - a| < \delta_1$ then $|f(x) - L| < 1$. Therefore, we have that

$$|f(x)| = |f(x) - L + L| \leq |f(x) - L| + |L| < 1 + |L|.$$

Let $K = 1 + |L|$, so that $|f(x)| < K$.

Now let $\epsilon = \frac{\epsilon}{2}$. Then, by definition, there exists $\delta_2 > 0$ such that if $x \in A$ and $|x - a| < \delta_2$ then $|g(x) - M| < \frac{\epsilon}{2K}$, and there exists $\delta_3 > 0$ such that if $x \in A$ and $|x - a| < \delta_3$ then $|f(x) - L| < \frac{\epsilon}{2|M| + 1}$.

Let $\delta = \min(\delta_1, \delta_2, \delta_3)$. Then, if $x \in A$ and $|x - a| < \delta$, we add and subtract $Mf(x)$ to get

$$\begin{aligned} |(fg)(x) - LM| &= |f(x)g(x) - Mf(x) + Mf(x) - LM| \\ &= |f(x)(g(x) - M) + M(f(x) - L)| \\ &\leq |f(x)| \cdot |g(x) - M| + |M| \cdot |f(x) - L| \\ &< K \cdot \frac{\epsilon}{2K} + |M| \cdot \frac{\epsilon}{2|M| + 1} \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon. \end{aligned}$$

Therefore, as $h(x) = (fg)(x)$, we have that $|h(x) - (L \cdot M)| < \epsilon$. This suffices to prove that $\lim_{x \rightarrow a} (fg)(x) = L \cdot M$. $\square$

Proof of Theorem 11.9(c). Let $\lim_{x \rightarrow a} f(x) = L$ where $L \neq 0$. Then, suppose there exists a function $h(x) = \frac{1}{f}(x)$. To show that $\lim_{x \rightarrow a} \frac{1}{f}(x) = \frac{1}{L}$, it suffices to show that $|h(x) - \frac{1}{L}| < \epsilon$.

As $\lim_{x \rightarrow a} f(x) = L$, we have that for every $\epsilon > 0$, there exists $\delta_1 > 0$ such that if $x \in A$ and $0 < |x - a| < \delta_1$ then by **Lemma 11.8**, $|f(x)| > \frac{|L|}{2}$. Therefore, we have that

$$\left| \frac{1}{f(x)} - \frac{1}{L} \right| = \left| \frac{L - f(x)}{f(x) \cdot L} \right| = \frac{|f(x) - L|}{|f(x)| \cdot |L|} < \frac{|f(x) - L|}{\frac{|L|}{2} \cdot |L|} = \frac{2|f(x) - L|}{L^2}.$$

Now let $\epsilon = \frac{\epsilon}{2}$. Then, since $\frac{\epsilon \cdot L^2}{2} > 0$, there exists $\delta_2 > 0$ such that if $x \in A$ and $0 < |x - a| < \delta_2$ then $|f(x) - L| < \frac{\epsilon \cdot L^2}{2}$.

Let $\delta = \min(\delta_1, \delta_2)$. Then, if $x \in A$ and $0 < |x - a| < \delta$, we have that

$$\left| \frac{1}{f(x)} - \frac{1}{L} \right| < \frac{2|f(x) - L|}{L^2} < \frac{2 \cdot \frac{\epsilon \cdot L^2}{2}}{L^2} = \epsilon.$$

Therefore, as $h(x) = \frac{1}{f}(x)$, we have that $|h(x) - \frac{1}{L}| < \epsilon$. This suffices to prove that $\lim_{x \rightarrow a} \frac{1}{f}(x) = \frac{1}{L}$. □

Corollary 11.10. *If f and g are continuous at a , then $f + g$ and fg are continuous at a . Also, $\frac{1}{f}$ and $\frac{g}{f}$ are continuous at a , provided that $f(a) \neq 0$.*

Proof. First, we want to show that $\lim_{x \rightarrow a} (f+g)(x) = (f+g)(a)$. We know that f, g are continuous at a , hence $\lim_{x \rightarrow a} f(x) = f(a)$ and $\lim_{x \rightarrow a} g(x) = g(a)$. Then, by Theorem 11.9,

$$\lim_{x \rightarrow a} (f+g)(x) = f(a) + g(a) = (f+g)(a).$$

Second, we want to show that $\lim_{x \rightarrow a} (fg)(x) = (fg)(a)$. We know that f, g are continuous at a , hence $\lim_{x \rightarrow a} f(x) = f(a)$ and $\lim_{x \rightarrow a} g(x) = g(a)$. Then, by Theorem 11.9,

$$\lim_{x \rightarrow a} (fg)(x) = f(a) \cdot g(a) = (fg)(a).$$

Third, we want to show that $\lim_{x \rightarrow a} \frac{1}{f}(x) = \frac{1}{f}(a)$. Since f is continuous at a and $f(a) \neq 0$, by continuity there exists a neighbourhood $(a - \delta, a + \delta)$ on which $f(x) \neq 0$, so $\frac{1}{f}$ is well-defined in this neighbourhood and the limit is meaningful. We have $\lim_{x \rightarrow a} f(x) = f(a) \neq 0$, so by Theorem 11.9,

$$\lim_{x \rightarrow a} \frac{1}{f}(x) = \frac{1}{f(a)} = \frac{1}{f}(a).$$

Fourth, we want to show that $\lim_{x \rightarrow a} \frac{g}{f}(x) = \frac{g}{f}(a)$. Since $f(a) \neq 0$, the function $\frac{1}{f}$ is well-defined and continuous at a by the third part of this proof. We write $\frac{g}{f} = g \cdot \frac{1}{f}$, and since both g and $\frac{1}{f}$ are continuous at a , we have $\lim_{x \rightarrow a} g(x) = g(a)$ and $\lim_{x \rightarrow a} \frac{1}{f}(x) = \frac{1}{f}(a)$. Then by Theorem 11.9,

$$\lim_{x \rightarrow a} \frac{g}{f}(x) = g(a) \cdot \frac{1}{f(a)} = \frac{g}{f}(a).$$

This completes the proof. □

Definition 11.11. A polynomial in one variable with real coefficients is a function f of the form

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

for some $n \in \mathbb{N} \cup \{0\}$, where $a_i \in \mathbb{R}$ for $0 \leq i \leq n$. A rational function in one variable with real coefficients is a function of the form $h(x) = \frac{f(x)}{g(x)}$ where f and g are polynomials in one variable with real coefficients.

Corollary 11.12. Polynomials in one variable with real coefficients are continuous. A rational function in one variable with real coefficients $h(x) = \frac{f(x)}{g(x)}$ is continuous at all $a \in \mathbb{R}$ where $g(a) \neq 0$.

Proof of Corollary 11.12. Let $f(x) = \sum_{i=0}^n a_i x^i$ for some $n \in \mathbb{N} \cup \{0\}$ where $a_i \in \mathbb{R}$ for $0 \leq i \leq n$. We want to show that $f(x)$ is continuous.

Firstly, we want to show that x^n is continuous for all $n \in \mathbb{N} \cup \{0\}$. We show this by induction on n . For the base case, $n = 0$, we have that $x^0 = 1$ is continuous as it is affine. For the inductive step, we have that $x^{n+1} = x^n \cdot x$ is continuous by **Corollary 11.10**, since x^n is continuous by the inductive hypothesis and x is continuous as it is affine.

Secondly, we want to show that $a_i x^i$ is continuous for all $n \in \mathbb{N} \cup \{0\}$. We have that a_i is affine and therefore continuous, and x^i is continuous as shown above. Hence, it follows by **Corollary 11.10** that $a_i x^i$ is continuous.

Now we want to show that $f(x) = \sum_{i=0}^n a_i x^i$ is continuous. We show this by induction on n . For the base case, $n = 0$, we have that $f(x) = a_0$ is affine and therefore continuous. For the inductive step, we can write

$$f(x) = \sum_{i=0}^{n+1} a_i x^i = a_{n+1} x^{n+1} + \sum_{i=0}^n a_i x^i.$$

We have that $a_{n+1} x^{n+1}$ is continuous as shown above, and $\sum_{i=0}^n a_i x^i$ is continuous by the inductive hypothesis. Therefore, it follows by **Corollary 11.10** that $f(x) = \sum_{i=0}^{n+1} a_i x^i$ is continuous.

Therefore, all polynomials are continuous. □

Now we want to look at limits of the composition of functions. We assume here (for 11.13 - 11.15) that $a \in A$, $g: A \rightarrow \mathbb{R}$ and $f: I \rightarrow \mathbb{R}$, where I is an open interval containing $\overline{g(A)}$. It is not quite true in general that if $\lim_{x \rightarrow a} g(x) = M$ and $\lim_{y \rightarrow M} f(y) = L$, then $\lim_{x \rightarrow a} f(g(x)) = L$, but it is true in some cases.

Theorem 11.13. If $\lim_{x \rightarrow a} g(x) = M$ and f is continuous at M , then $\lim_{x \rightarrow a} f(g(x)) = f(M)$.

Proof of Theorem 11.13. Let $\lim_{x \rightarrow a} g(x) = M$ and f is continuous at M , then we want to show that $\lim_{x \rightarrow a} f(g(x)) = f(M)$.

As f is continuous at M , we can say that for every $\epsilon > 0$, there exists $\delta_1 > 0$ such that if $y \in I$ and $|y - M| < \delta$ then $|f(y) - f(M)| < \epsilon$.

Now consider $\lim_{x \rightarrow a} g(x) = M$. We have that, then, for all $\epsilon > 0$ there exists $\delta_2 > 0$ such that $|y - a| < \delta_2$. As $y \in I$ and I is an open interval containing $g(a)$, then, we can rewrite $|y - M| < \delta$ as $|g(x) - M| < \delta_1$.

Then, it clearly follows that as $y = g(x)$, we get that $|g(x) - M| < \delta_1$. From this, we can say $|f(g(x)) - f(M)| < \epsilon$ which shows that $\lim_{x \rightarrow a} f(g(x)) = f(M)$. This suffices to complete the proof. $\square$

Remark 11.14. This theorem can also be rewritten as

$$\lim_{x \rightarrow a} f(g(x)) = f\left(\lim_{x \rightarrow a} g(x)\right),$$

which can be remembered as “limits pass through continuous functions.”

Corollary 11.15. *If g is continuous at a and f is continuous at $g(a)$, then $f \circ g$ is continuous at a .*

Proof of Corollary 11.15. Let g be continuous at a and f be continuous at $g(a)$. We want to show that $f \circ g$ is continuous at a .

As g is continuous at a we can say that for all $\epsilon > 0$, there exists $\delta > 0$ such that if $y \in A$ and $|y - a| < \delta_1$ then $|f(y) - f(a)| < \epsilon$. Then, as f is continuous at $g(a)$, there exists $\delta_2 > 0$ such that if $g(x) \in I$ and $|g(x) - g(a)| < \delta_2$ then $|f(g(a)) - f(g(x))| < \epsilon$.

Now consider $h(x) = f(g(x)) = f \circ g$. To show that $h(x)$ is continuous at a , it suffices to show that for every $\epsilon > 0$, there exists $\delta > 0$ such that if $x \in A$ and $|x - a| < \delta$ then $|h(x) - h(a)| < \epsilon$. From **Theorem 11.13**, we have that if $\lim_{x \rightarrow a} g(x) = M$ and f is continuous at M , then $\lim_{x \rightarrow a} f(g(x)) = f(M)$. As, by **Theorem 11.5**, we have that $\lim_{x \rightarrow a} g(x) = M$ is equivalent to saying g is continuous at a , we have that that $f \circ g$ is continuous at a . $\square$

Additional Exercises

1. Find the following limits, stating clearly which results you are using.

a) $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$.

b) $\lim_{h \rightarrow 0} \frac{\sqrt{a+h} - \sqrt{a}}{h}$.

Additional Exercise 1(a). We want to find the limit of $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$.. Consider the expression $x^3 - 8$. By **Defintiiion 11.11** , we know that we can write this formula as a polynomial in one variable. Then, we can rewrite this expression as $(x - 2)(x^2 + 2x + 4)$

Now, consider $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$. We can write this as $\lim_{x \rightarrow 2} \frac{(x-2)(x^2+2x+4)}{x-2}$. We can simplify this to get $\lim_{x \rightarrow 2} (x^2 + 2x + 4)$ where we get 12. $\square$

Additional Exercise 1(b). We want to find the limit of $\lim_{h \rightarrow 0} \frac{\sqrt{a+h} - \sqrt{a}}{h}$.

Consider $\lim_{h \rightarrow 0} \frac{\sqrt{a+h} - \sqrt{a}}{h} \cdot \frac{\sqrt{a+h} + \sqrt{a}}{\sqrt{a+h} + \sqrt{a}}$. We have that

$$\frac{\sqrt{a+h} - \sqrt{a}}{h} \cdot \frac{\sqrt{a+h} + \sqrt{a}}{\sqrt{a+h} + \sqrt{a}} = \frac{(a+h) - a}{h(\sqrt{a+h} + \sqrt{a})} = \frac{h}{h(\sqrt{a+h} + \sqrt{a})} = \frac{1}{\sqrt{a+h} + \sqrt{a}}.$$

Therefore, we have that

$$\lim_{h \rightarrow 0} \frac{\sqrt{a+h} - \sqrt{a}}{h} = \lim_{h \rightarrow 0} \frac{1}{\sqrt{a+h} + \sqrt{a}}.$$

Now, as $h \rightarrow 0$, we have that $\sqrt{a+h} \rightarrow \sqrt{a}$, and therefore $\sqrt{a+h} + \sqrt{a} \rightarrow 2\sqrt{a}$. Hence, it follows that

$$\lim_{h \rightarrow 0} \frac{\sqrt{a+h} - \sqrt{a}}{h} = \frac{1}{2\sqrt{a}}.$$

Therefore, we have that $\lim_{h \rightarrow 0} \frac{\sqrt{a+h} - \sqrt{a}}{h} = \frac{1}{2\sqrt{a}}$. $\square$

2. Let $A \subset \mathbb{R}$ be an open set, $f: A \rightarrow \mathbb{R}$, and $a \in A$. Since A is open there is a region $R \subset A$ containing a . Prove that

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} f|_R(x).$$

(Note that this should be interpreted as meaning that one limit exists if, and only if, the other limit exists, and if the limits exist, they are equal.)

3. (*The Sandwich/Squeeze Theorem*) Let $A \subset \mathbb{R}$ be an open set. Let $a \in A$. Suppose that $f, g, h: A \rightarrow \mathbb{R}$ satisfy

$$f(x) \leq g(x) \leq h(x), \quad \text{for all } x \in A,$$

and $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} h(x)$ both exist and equal L . Then $\lim_{x \rightarrow a} g(x)$ also exists and equals L .

Additional Exercise of 3. Let $A \subset \mathbb{R}$ be an open set and let $a \in A$. Suppose that $f, g, h: A \rightarrow \mathbb{R}$ satisfy $f(x) \leq g(x) \leq h(x)$ for all $x \in A$, and that $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} h(x) = L$. We want to show that $\lim_{x \rightarrow a} g(x) = L$.

Let $\epsilon > 0$. Since $\lim_{x \rightarrow a} f(x) = L$, there exists $\delta_1 > 0$ such that for all $x \in A$ with $0 < |x - a| < \delta_1$ we get that $|f(x) - L| < \epsilon$. We can rewrite this expression as $L - \epsilon < f(x) < L + \epsilon$.

Now consider $h(x)$. Since $\lim_{x \rightarrow a} h(x) = L$, there exists $\delta_2 > 0$ such that for all $x \in A$ with $0 < |x - a| < \delta_2$ we get that $|h(x) - L| < \epsilon$. We can rewrite this expression as $L - \epsilon < h(x) < L + \epsilon$.

Let $\delta = \min(\delta_1, \delta_2)$. Then for all $x \in A$ with $0 < |x - a| < \delta$, we have both inequalities simultaneously, and since $f(x) \leq g(x) \leq h(x)$:

$$L - \epsilon < f(x) \leq g(x) \leq h(x) < L + \epsilon.$$

Therefore $|g(x) - L| < \epsilon$. Since $\epsilon > 0$ was arbitrary, this suffices to prove that $\lim_{x \rightarrow a} g(x) = L$. □

4. Let $f: A \rightarrow \mathbb{R}$ be a real-valued function with domain $A \subset \mathbb{R}$, A an open set. Let $a \in \mathbb{R}$ and let $b \in \mathbb{R}$ be such that $(a, b) \subset A$. The *right-hand limit* of f at a is a real number L satisfying the following condition: for all $\epsilon > 0$, there is some $\delta > 0$ such that if $x \in A$ and $0 < x - a < \delta$, then $|f(x) - L| < \epsilon$. If such an L exists we write this as

$$\lim_{x \rightarrow a^+} f(x) = L.$$

- a) Formulate an analogous definition for the limit from below/left-hand limit of f at b , written $\lim_{x \rightarrow b^-} f(x)$.
- b) Let $x_0 \in A$. Prove that $\lim_{x \rightarrow x_0} f(x)$ exists and equals L if, and only if, $\lim_{x \rightarrow x_0^+} f(x)$ and $\lim_{x \rightarrow x_0^-} f(x)$ both exist and equal L .
5. (a) Show that $f(x) = \sqrt{x}$ is continuous at $x = 1$.
 (b) Show that $f(x) = x^2$ is continuous at $x = 2$.
6. Consider functions $f, g, h : \mathbb{R} \rightarrow \mathbb{R}$ such that, for all $x \neq 1$,
- $$|f(x) - 2| < |x - 1|^2, \quad |g(x) - 3| < 2|x - 1|, \quad |h(x) - 5| < |x - 1|.$$
- (a) Find a $\delta > 0$ such that if $0 < |x - 1| < \delta$, then $|f(x)g(x) - h(x) - 1| < \frac{1}{1000}$.
 (b) Prove that $\lim_{x \rightarrow 1} (f(x)g(x) - h(x)) = 1$.

7. (From Spivaks' Calculus) In each of the following, justify your answer fully.

- a) If $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ do not exist, can $\lim_{x \rightarrow a} [f(x) + g(x)]$ exist? Can $\lim_{x \rightarrow a} [f(x)g(x)]$ exist?
- b) If $\lim_{x \rightarrow a} f(x)$ exists and $\lim_{x \rightarrow a} [f(x) + g(x)]$ exists, must $\lim_{x \rightarrow a} g(x)$ exist?
- c) If $\lim_{x \rightarrow a} f(x)$ exists and $\lim_{x \rightarrow a} [f(x)g(x)]$ exists, does it follow that $\lim_{x \rightarrow a} g(x)$ exists?

Additional Exercise 7(a). **Firstly**, we want to show that if $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ do not exist, $\lim_{x \rightarrow a} [f(x) + g(x)]$ can still exist.

Let $f(x) = \frac{1}{x}$ and $g(x) = -\frac{1}{x}$. Consider $a = 0$. Then, we have that $\lim_{x \rightarrow 0} f(x)$ and $\lim_{x \rightarrow 0} g(x)$ do not exist.

Consider the function $h(x) = f(x) + g(x)$. By the definition of the functions, we have that $h(x) = \frac{1}{x} - \frac{1}{x} = 0$ for all $x \neq 0$, and therefore $\lim_{x \rightarrow 0} [f(x) + g(x)] = 0$. Thus, the limit of the sum exists even though the individual limits do not.

Secondly we want to show that if $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ do not exist, $\lim_{x \rightarrow a} [f(x)g(x)]$ can still exist.

$$\text{Let } f(x) = g(x) = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \end{cases}.$$

Then, we have that $\lim_{x \rightarrow 0} f(x)$ and $\lim_{x \rightarrow 0} g(x)$ do not exist. However, we have that $f(x)g(x) = 1$ for all $x \neq 0$, and therefore $\lim_{x \rightarrow 0} [f(x)g(x)] = 1$. Thus, the limit of the product exists even though the individual limits do not.

□

Additional Exercise 7b. Let $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} [f(x) + g(x)]$ exists. We want to show that $\lim_{x \rightarrow a} g(x)$ must exist.

We know that $\lim_{x \rightarrow a} [f(x) + g(x)]$ exists. We can write this as $\lim_{x \rightarrow a} (f + g)(x) = L + M$. Suppose $\lim_{x \rightarrow a} f(x) = L$. Then, we have that $\lim_{x \rightarrow a} (f + g)(x) - \lim_{x \rightarrow a} f(x) = L + M - L = M$.

Now consider $g(x)$. We can say that $g(x) = [f(x) + g(x)] - f(x)$. We can take the limits of both sides to get

$$\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} [f(x) + g(x)] - \lim_{x \rightarrow a} f(x)$$

.

We have that $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} (f + g)(x) = L + M$. Therefore, we can rewrite the equation to get

$$\lim_{x \rightarrow a} g(x) = L + M - L = M$$

Hence, we have that $\lim_{x \rightarrow a} g(x) = M$.

□

Additional Exercise 7(c). Let $\lim_{x \rightarrow a} f(x)$ exists and $\lim_{x \rightarrow a} [f(x)g(x)]$ exists. We want to show that it is not necessary that $\lim_{x \rightarrow a} g(x)$ exists. We will prove this through counterexample.

Let $a = 0$, $f(x) = 0$, and $g(x) = \frac{1}{x}$. Then, consider the function $f(x)g(x) = 0 \cdot \frac{1}{x} = 0$. We have that $\lim_{x \rightarrow 0} f(x)$ exists and $\lim_{x \rightarrow 0} [f(x)g(x)]$ exists. Though, we know that $\lim_{x \rightarrow 0} g(x)$ does not exist when $a = 0$.

Therefore, this suffices to prove the statement that $\lim_{x \rightarrow a} f(x)$ exists and $\lim_{x \rightarrow a} [f(x)g(x)]$ exists but $\lim_{x \rightarrow a} g(x)$ may not necessarily exist.

□

8. Give an example to show that $\lim_{x \rightarrow a} f(g(x)) = f(\lim_{x \rightarrow a} g(x))$ need not hold when f is not continuous.

9. Define the function $f : \mathbb{R} \rightarrow \mathbb{R}$ as

$$f(x) := \begin{cases} 0, & \text{if } x \in \mathbb{R} \setminus \mathbb{Q}, \\ \frac{1}{q}, & \text{if } x = \frac{p}{q} \text{ where } q \in \mathbb{N} \text{ and } \gcd(p, q) = 1, \\ 1, & \text{if } x = 0. \end{cases}$$

(Note that $\gcd(p, q) = 1$ implies that $x = p/q$ is in its most reduced form.) Where is f continuous?

10. Consider an increasing function $f : \mathbb{R} \rightarrow \mathbb{R}$. Prove that the set of points, where f is not continuous, is countable.